

Non-Standard Interactions (NSI) in tpeanuts

Fast Simulation of Neutrino Oscillations in Matter

MSc Thesis (Trabajo Fin de Máster) — BSM Extension Companion Document

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This document is the companion reference for the `core.BSM.bsm_nsi` package and every downstream consumer of `OscillationParameters.nsi`: the exact-Hamiltonian builders in `core.common.hamiltonian`, the numerical evolver (`core.numerical.evolver`), the perturbative segment evolver (`core.perturbative.evolver`), and the solar/Earth/atmosphere propagation pipelines (`medium.solar`, `medium.earth`, `medium.atmosphere`). Chapter 1 covers the theoretical foundations of propagation NSI, including the electron/proton-versus-neutron composition dependence of the effective coupling. Chapter 2 is an exhaustive account of the `tpeanuts` implementation, with every mathematical relation used in the code. Chapter 3 documents the named NSI presets and the observables/simulations that can be built with them. Chapter 4 walks through the validation and inference notebooks that exercise NSI and reports their actual results. Blue boxes (**Theoretical Foundation**) contain the physics; green boxes (**Implementation**) link that physics to actual `tpeanuts` code; orange boxes collect approximations and limitations; cyan/purple boxes (**Result**) report a concrete numerical outcome of a notebook or of this project's own test suite. Text highlighted in yellow marks key results/equations.

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Theoretical Foundations of Non-Standard Interactions

This chapter develops the physics of propagation Non-Standard Interactions (NSI) from the coherent-forward-scattering matter potential up to the fermion-composition dependence of the effective coupling matrix. It is deliberately implementation-agnostic: [Chapter 2](#) maps every equation derived here onto the concrete `tpeanuts` code that evaluates it.

1.1 The Standard Model matter potential

Theoretical Foundation

A neutrino propagating through ordinary matter undergoes coherent forward scattering off the electrons, protons, and neutrons of the medium. For ν_e , an additional charged-current (CC) diagram (via W exchange with electrons) is possible on top of the neutral-current (NC) diagram common to all three active flavours, giving the Wolfenstein matter potential [13, 16]

$$V_{CC} = \pm\sqrt{2} G_F n_e, \quad (1.1)$$

with the sign fixed by the neutrino/antineutrino convention (+ for neutrinos, – for antineutrinos). In the flavour basis (ν_e, ν_μ, ν_τ) this contributes $\text{diag}(V_{CC}, 0, 0)$ to the propagation Hamiltonian: the NC piece is common to all three flavours and is therefore an unobservable global phase in the pure Standard Model, 3-flavour case (it becomes observable only once a fourth, NC-blind sterile state is introduced – a separate BSM extension, not covered by this document).

1.2 Non-Standard Interactions: the effective ϵ matrix

Theoretical Foundation

NSI generalise the coherent-forward-scattering picture by allowing new, model-independent four-fermion operators of the form

$$\mathcal{L}_{\text{NSI}} = -2\sqrt{2} G_F \epsilon_{\alpha\beta}^{f,P} (\bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta) (\bar{f} \gamma_\mu P f), \quad P \in \{L, R\}, \quad (1.2)$$

where f is a matter fermion (electron, up quark, or down quark) and $\epsilon_{\alpha\beta}^{f,P}$ is a dimensionless coupling, generally complex for $\alpha \neq \beta$ [8]. When a neutrino of flavour α coherently forward-scatters off a medium containing many such fermions, the sum over f and P collapses into a single effective coupling per flavour pair, and the propagation Hamiltonian

in the flavour basis becomes

$$H_{\text{mat}}^{\text{NSI}} = V_{CC} \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \epsilon \right), \quad (1.3)$$

with ϵ a dimensionless 3×3 *Hermitian* matrix,

$$\epsilon = \begin{pmatrix} \epsilon_{ee} & \epsilon_{e\mu} & \epsilon_{e\tau} \\ \epsilon_{e\mu}^* & \epsilon_{\mu\mu} & \epsilon_{\mu\tau} \\ \epsilon_{e\tau}^* & \epsilon_{\mu\tau}^* & \epsilon_{\tau\tau} \end{pmatrix}. \quad (1.4)$$

Diagonal entries are real (Hermiticity); off-diagonal entries are generally complex, each carrying its own CP-violating phase independent of the standard PMNS δ . The Standard Model limit is $\epsilon = 0$.

Hermiticity of ϵ is not a modelling choice: it is required for $H_{\text{mat}}^{\text{NSI}}$ itself to be Hermitian, and hence for the resulting time evolution to be unitary (probability-conserving). Only the traceless part of ϵ is observable in oscillation probabilities – adding cI_3 to ϵ for any real c shifts $H_{\text{mat}}^{\text{NSI}}$ by $cV_{CC}I_3$, and because I_3 commutes with every other term in the Hamiltonian at every point along the trajectory, this generalised-time-ordered exponential factorises into an overall, position-dependent but flavour-independent phase, exactly as for the plain NC term above. Fixing $\epsilon_{\mu\mu} = 0$ is therefore a valid convention for the diagonal sector, not a loss of generality.

Approximations and Limitations

Antineutrinos. The matter background is not its own antimatter, so the effective potential seen by an antineutrino uses the complex-conjugated matrix, $\epsilon \rightarrow \epsilon^*$, on top of the usual $V_{CC} \rightarrow -V_{CC}$ sign flip – exactly mirroring the $U \rightarrow U^*$ convention already used for the PMNS mixing matrix itself. Off-diagonal CP-violating phases therefore flip sign for antineutrinos, as physically required.

1.3 Fermion decomposition and the electron/proton-versus-neutron split

Theoretical Foundation

The single matrix ϵ of Eq. (1.4) is not itself a fundamental object: it is a matter-composition-weighted combination of three independent, per-fermion couplings ϵ^e , ϵ^u , ϵ^d [6, 12, 14]. Writing $n_e(x)$, $n_p(x)$, $n_n(x)$ for the local electron, proton, and neutron number densities, the effective coupling actually entering the Hamiltonian at position x is

$$\epsilon_{\alpha\beta}(x) = \epsilon_{\alpha\beta}^e + \frac{n_p(x)}{n_e(x)} \epsilon_{\alpha\beta}^p + \frac{n_n(x)}{n_e(x)} \epsilon_{\alpha\beta}^n, \quad \epsilon^p \equiv 2\epsilon^u + \epsilon^d, \quad \epsilon^n \equiv \epsilon^u + 2\epsilon^d, \quad (1.5)$$

where ϵ^p/ϵ^n are themselves fixed linear combinations of the up- and down-quark couplings (a proton is uud , a neutron udd). Ordinary matter is electrically neutral, $n_p(x) \equiv n_e(x)$

at every point, so the three-fermion sum collapses to exactly *two* effective pieces rather than three independent physical ones:

$$\epsilon_{\alpha\beta}(x) = \epsilon_{\alpha\beta}^{(e+p)} + \frac{n_n(x)}{n_e(x)} \epsilon_{\alpha\beta}^n, \quad \epsilon^{(e+p)} \equiv \epsilon^e + 2\epsilon^u + \epsilon^d. \quad (1.6)$$

The label “electron/proton-normalized” for $\epsilon^{(e+p)}$ is therefore a statement about which density it multiplies ($n_e(x)$, via $n_p(x) \equiv n_e(x)$), not a claim that it is itself a single fundamental coupling – it is the fixed combination $\epsilon^e + 2\epsilon^u + \epsilon^d$ that happens to be proportional to $n_e(x)$ in any electrically neutral medium.

Approximations and Limitations

Published bounds are single-composition fits, not fundamental couplings.

Every published NSI bound or best-fit value – including every preset in [Chapter 3](#) – is a measurement of the *effective*, single-medium-composition ϵ of [Eq. \(1.4\)](#), typically for Earth-like matter ($n_n(x)/n_e(x) \approx 1$), not of ϵ^e , ϵ^u , ϵ^d individually. Setting $\epsilon^n = 0$ (i.e. using only [Eq. \(1.4\)](#), without [Eq. \(1.6\)](#)’s second term) is therefore a *modelling hypothesis* carried by whichever preset or analysis quotes the bound, not a physical fact derivable from the quoted number: two different experiments with different target composition can each report an “ ϵ_{ee} ” consistent with a common $(\epsilon^e, \epsilon^u, \epsilon^d)$ but numerically different $\epsilon^{(e+p)}$, and neither measurement alone fixes ϵ^n . This composition dependence is also the reason the LMA-Dark degenerate solution ([Section 1.5](#)) is sometimes called into question when different target compositions are combined in a single global fit [\[3\]](#).

Theoretical Foundation

Magnitude of the composition dependence. $n_n(x)/n_e(x)$ is not a universal number – it is a genuine function of position within a given medium, and it varies sharply between media relevant to **tpeanuts**:

- **The Sun.** The B16 standard solar model [\[15\]](#) gives a hydrogen-dominated envelope where $n_n(r)/n_e(r) \rightarrow 0$ (a proton has no neutrons to contribute), rising smoothly to $n_n(r)/n_e(r) \approx 1$ in the helium/metal-enriched core, where a non-negligible fraction of nucleons are bound in ${}^4\text{He}$ and heavier nuclei with roughly equal proton and neutron content.
- **The Earth.** The Preliminary Reference Earth Model (PREM) [\[4\]](#) gives $n_n(x)/n_e(x)$ in the range ~ 1.0 – 1.15 across the mantle/core boundary, reflecting the modest neutron excess of common terrestrial nuclei (silicates in the mantle, iron in the core) relative to a strictly isoscalar medium.

A single constant ϵ (i.e. $\epsilon^n = 0$) is therefore an excellent approximation for propagation confined to one roughly isoscalar medium – which is why it was the only term implemented for most of this project’s history – but is not exact across media of very different composition, and is a comparatively crude approximation for propagation through the Sun’s envelope specifically, where $n_n(r)/n_e(r)$ swings from ≈ 0 to ≈ 1 over the trajectory.

1.4 The 3+1 sterile extension and its neutral-current term

Theoretical Foundation

NSI can be combined freely with the independent 3+1 sterile-neutrino extension (a fourth, gauge-singlet mass eigenstate with its own mixing angles $\theta_{14}, \theta_{24}, \theta_{34}$ and mass splitting Δm_{41}^2). Because the sterile state is an $SU(2)_L \times U(1)_Y$ singlet, it does not receive the V_{CC} term, but it also does not receive the neutral-current potential $V_{NC} = \mp \frac{1}{\sqrt{2}} G_F n_n$ that the three active flavours share equally. In the pure 3-flavour case that shared V_{NC} is an unobservable common phase (§1.1); once a sterile state is present, subtracting it as an overall $V_{NC} I_4$ phase leaves a genuine relative potential on the sterile diagonal entry,

$$\text{diag}(V_{CC} + V_{NC}, V_{NC}, V_{NC}, 0) - V_{NC} I_4 = \text{diag}(V_{CC}, 0, 0, -V_{NC}). \quad (1.7)$$

This sterile neutral-current (NC) term is conceptually and mathematically independent of the NSI composition term of Eq. (1.6): both need the local neutron density $n_n(x)$, but they are additive, independently switchable contributions (Chapter 2), and the sterile NC term applies only for a 4-flavour scenario, while the NSI composition term applies for any flavour count.

1.5 The LMA-Dark degeneracy

Theoretical Foundation

The standard LMA solar solution ($\theta_{12} \approx 33.4^\circ$) admits a discrete degeneracy in global oscillation data: replacing $\theta_{12} \rightarrow \pi/2 - \theta_{12}$ ($\approx 56.6^\circ$) together with

$$\epsilon_{ee} \rightarrow -(2 + \epsilon_{ee}) \approx -2.0 \quad (1.8)$$

reproduces an equivalent fit to solar and reactor data (LMA-Dark; [5]). The large negative ϵ_{ee} effectively flips the sign of the ν_e matter potential, so that the octant-flipped mixing angle produces the same resonance structure as the standard solution with the standard sign. This is a striking illustration that a sufficiently large NSI coupling can hide an entire discrete ambiguity in the standard oscillation parameters, and is the physics case validated numerically in Chapter 4.

Key References

Original NSI proposal: [8]. Model-independent one-parameter bounds: [2]. Per-fermion decomposition and matter composition dependence: [6, 12, 14]. LMA-Dark degeneracy and its composition sensitivity: [3, 5]. Standard matter potential: [13, 16]. Solar and Earth composition profiles: [4, 15].

Chapter 2

Implementation in tpeanuts

This chapter maps every equation of [Chapter 1](#) onto the concrete `tpeanuts` implementation, module by module. There is no separate `core.BSM` Hamiltonian-assembly module: `hamiltonian_reduced` and `hamiltonian_flavour` in `core.common.hamiltonian` already handle NSI, the 3+1 sterile extension, or both simultaneously, dispatching purely on which fields of `OscillationParameters` are populated.

2.1 NSIConfig: the parameter container

`core/BSM/bsm_nsi.py`

frozen dataclass storing every NSI coupling and building the corresponding complex tensors.

Implementation in tpeanuts

NSIConfig stores the electron/proton-normalized block ϵ as nine real scalar fields (three real diagonal entries plus three complex off-diagonal entries decomposed into real/imaginary parts: `eps_ee`, `eps_mumu`, `eps_tautau`, `eps_emu_re/_im`, `eps_etau_re/_im`, `eps_mutau_re/_im`), and the neutron-normalized block ϵ^n ([Eq. \(1.6\)](#)) as the mirrored nine fields `eps_ee_n`, ..., `eps_mutau_n_im`. Every field defaults to 0.0, so $\epsilon^n = 0$ (no composition dependence) reproduces every pre-existing single-matrix scenario exactly. `__post_init__` recomputes *both* derived complex 3×3 tensors, `epsilon` and `epsilon_n`, from their respective nine scalar fields on *every* construction – including every `dataclasses.replace(...)` call, since that re-invokes `__init__`/`__post_init__`. This makes the eighteen scalar fields the single source of truth: tweaking one field via `replace` can never leave a stale `epsilon`/`epsilon_n` behind.

Differentiable tensor construction

Both 3×3 Hermitian matrices are assembled by the shared helper `_hermitian_3x3`, built entirely from torch operations (`torch.complex`, `torch.stack`) rather than Python's `complex()` builtin or `torch.tensor(nested_python_list)`:

```
def _hermitian_3x3(ee, mumu, tautau, emu_re, emu_im,
                  etau_re, etau_im, mutau_re, mutau_im,
                  *, device, real_dtype):
    def _rt(x):
        return as_tensor(x, dtype=real_dtype, device=resolved_device)

    zero = torch.zeros((), dtype=real_dtype, device=resolved_device)
    ee_c, mumu_c, tautau_c = (torch.complex(_rt(v), zero)
                              for v in (ee, mumu, tautau))
    emu_c = torch.complex(_rt(emu_re), _rt(emu_im))
    etau_c = torch.complex(_rt(etau_re), _rt(etau_im))
    mutau_c = torch.complex(_rt(mutau_re), _rt(mutau_im))

    row0 = torch.stack((ee_c, emu_c, etau_c))
    row1 = torch.stack((emu_c.conj(), mumu_c, mutau_c))
    row2 = torch.stack((etau_c.conj(), mutau_c.conj(), tautau_c))
    return torch.stack((row0, row1, row2)).to(dtype=cdtype)
```

Every one of the eighteen scalar fields therefore accepts either a plain Python float (producing a non-grad-tracking leaf tensor, exactly as before) *or* a differentiable `torch.Tensor` with `requires_grad=True`, in which case the gradient reaches `epsilon/epsilon_n` intact. `device=None` resolves to CPU and every field is moved (differentiably, via `.to()`) onto that single resolved device, so mixing a GPU-resident coupling with plain-float ones never produces a mixed-device tensor. This is what lets `inference.solar_model.SolarNSIOscillationModel` (Section 2.7) fit `eps_ee` by gradient descent while building an ordinary `NSIConfig`, with no separate raw-tensor escape hatch needed for that case.

Approximations and Limitations

Identity-based equality. `NSIConfig` is declared with `@dataclass(frozen=True, eq=False)`: `__eq__`/`__hash__` are inherited from `object` (identity-based) rather than auto-generated from the fields. A field-tuple-based `__eq__` would be unsound here for two independent reasons: (i) comparing `torch.Tensor` fields with `==` does not reliably reduce to a plain boolean, and (ii) `NSIConfig.from_raw_epsilon` (below) leaves the eighteen scalar fields at their 0.0 defaults while setting `epsilon/epsilon_n` directly, so a field-tuple-based comparison would make a config carrying a non-zero raw ϵ compare equal to the plain Standard-Model-limit `NSIConfig()` – a real defect discovered and fixed during external review of this extension. Use `torch.equal(a.epsilon, b.epsilon)` directly for value comparison.

Preset construction and the raw-epsilon escape hatch

- `NSIConfig.from_preset(name)` builds a named configuration from `tpeanuts.config.presets.NSI_PRESETS` (Chapter 3).
- `NSIConfig.from_raw_epsilon(epsilon, *, epsilon_n=None)` is the escape hatch for an ϵ (and, optionally, ϵ^n) that does not decompose into the eighteen-scalar parametrization – e.g. a precomputed matrix. It validates that each tensor supplied has at least two dimensions, is square, is finite, and is Hermitian within a dtype-scaled tolerance (raising `ValueError` otherwise), and, when both are supplied, that ϵ and ϵ^n share the same active-flavour block size. **A config built this way must not be passed through `dataclasses.replace()`**: since its scalar fields stay at their 0.0 defaults, `__post_init__` would silently recompute `epsilon/epsilon_n` back to zero on the next `replace()` call, discarding the raw tensor (and any gradient connected to it) with no warning. Build a fresh config with `from_raw_epsilon` again instead.

Convenience properties

- `is_sm_limit`: True iff *both* `epsilon` and `epsilon_n` are exactly the zero matrix.
- `has_cp_violation`: True iff any off-diagonal entry of *either* matrix has a non-zero imaginary part.
- `has_neutron_coupling`: True iff `epsilon_n` is not exactly zero. This is the single flag every downstream consumer in this chapter uses to decide whether the composition term (Eq. (1.6)) needs to be evaluated at all – it costs nothing extra when

False (the default), so every pre-existing scenario is bit-identical to before this extension existed.

- `eps_emu/eps_etau/eps_mutau` (and their `_n` counterparts) return the corresponding entry as a plain Python `complex`, via a helper that detaches a tensor field first – these are non-differentiable display/convenience accessors, not part of the gradient path.

2.2 Embedding ϵ/ϵ^n for an arbitrary flavour count

Implementation in `tpeanuts`

`epsilon_tensor(n_flavours=...)` and its mirror `epsilon_n_tensor(n_flavours=...)` embed the 3×3 active-flavour block in the top-left corner of an $n_{\text{flavours}} \times n_{\text{flavours}}$ zero matrix when $n_{\text{flavours}} > 3$ (the 3+1 sterile case): the sterile row and column carry no NSI coupling of either kind, since the sterile state is an $SU(2)_L \times U(1)_Y$ singlet and NSI operators are built from Standard-Model-charged fermion currents. Both methods share a single embedding implementation (`_embed_active`), parametrised only by which of `self.epsilon/self.epsilon_n` to embed, so the two code paths cannot silently diverge.

2.3 Hamiltonian assembly: the three-piece decomposition

`core/common/hamiltonian.py`

`hamiltonian_matter_reduced` – the single function assembling the matter Hamiltonian for every scenario.

Theoretical Foundation

`hamiltonian_matter_reduced` builds the flavour-basis matter term as up to *three* independent, additive pieces, then rotates the whole sum into the reduced basis via $O^\dagger(\cdot)O$ (§1.1, §1.2):

$$H_{\text{mat}}^{\text{flavour}} = \underbrace{V_{CC}(n_e)(\text{diag}(1, 0, \dots, 0) + \epsilon)}_{\text{piece 1: NSI, } \epsilon} + \underbrace{V_{NC}(n_n) \text{diag}(0, \dots, 0, -1)}_{\text{piece 2: sterile NC}} + \underbrace{V_{CC}(n_n) \epsilon^n}_{\text{piece 3: NSI composition}}, \quad (2.1)$$

with $V_{CC}(n) \equiv \pm\sqrt{2}G_F n L_{\text{scale}}$ the same charged-current potential function of §1.1, evaluated once on the electron density and once (for pieces 2 and 3) on the neutron density, and $V_{NC}(n)$ the neutral-current potential of §1.3. Piece 3 is exactly the composition term of Eq. (1.6),

$$V_{CC}(n_n) \epsilon^n = L_{\text{scale}} \sqrt{2} G_F n_n \epsilon^n = \left(L_{\text{scale}} \sqrt{2} G_F n_e \right) \frac{n_n}{n_e} \epsilon^n = V_{CC}(n_e) \frac{n_n}{n_e} \epsilon^n, \quad (2.2)$$

so re-evaluating the *same* `matter_potential_cc` function on the neutron density instead of the electron density is algebraically identical to the ratio form of Eq. (1.6), without ever dividing by n_e – avoiding a division that would be singular in vacuum ($n_e = n_n = 0$).

Implementation in tpeanuts

- `include_nc = n_n_mol_cm3` is not `None` and `n_flavours == 4`: piece 2 is active only for the 3+1 sterile extension, and only when the caller supplies a neutron density; omitting it recovers the CC-only sterile Hamiltonian used throughout earlier phases of this project, exactly as before this extension existed.
- `needs_eps_n = oscillation.BSM_extension_NSI` and `oscillation.nsi.has_neutron_coupling`: piece 3 is active for *any* flavour count, driven purely by whether $\epsilon^n \neq 0$ – not by whether the sterile extension is also active.
- **Piece 3 is not optional once requested.** If `needs_eps_n` is `True` but `n_n_mol_cm3` was not supplied, the function raises `ValueError` rather than silently falling back to the ϵ -only Hamiltonian. This is a deliberate asymmetry with piece 2: omitting `n_n_mol_cm3` for the sterile NC term is itself a well-defined, physically exact choice (“CC-only sterile propagation”), whereas a non-zero ϵ^n is an explicit user request that the function must not silently downgrade.
- The direction matrices P_{cc} , P_{nc} , P_{ϵ^n} (the $O^\dagger(\cdot)O$ -rotated, V -independent counterparts of the three pieces above) are built once by the shared helper `_matter_direction_reduced`, reused identically by both the exact Hamiltonian builder here and the perturbative segment evolver (Section 2.4).

Approximations and Limitations

Because `hamiltonian_matter_reduced` is the single function every other builder in this chapter delegates to, this `ValueError` is a defense-in-depth guarantee: any current or future caller anywhere in the pipeline that forgets to supply the neutron density for a non-zero ϵ^n configuration fails loudly at the point of Hamiltonian assembly, rather than silently returning a physically incomplete result.

2.4 The perturbative segment evolver

`core/perturbative/evolver.py`

`evolver_perturbative_segment` – first-order perturbative evolution for one piecewise-polynomial density segment (used by `medium.earth` and `medium.atmosphere`).

Theoretical Foundation

For a segment over which the electron/neutron density varies, the zeroth-order evolver U_0 uses the segment-average potential; the first-order correction U_1 is built from the spectral projectors M_a of the average Hamiltonian and the oscillatory integral of the *residual* density (density minus its segment average) against the Hamiltonian eigenvalue differences,

$$U_1 = -i \sum_{a,b} I_{ab} (M_a P M_b), \quad I_{ab} = \int_{x_1}^{x_2} \delta n(x) e^{-i\lambda_a x + i\lambda_b x} dx, \quad (2.3)$$

where P is whichever position-independent direction matrix the term in question needs (Eq. (2.1)).

Implementation in tpeanuts

`evolutor_first_order` accepts three independent, optional direction matrices, mirroring Eq. (2.1) exactly: `P` (piece 1), `P_nc` (piece 2), and `P_eps_n` (piece 3). The key implementation observation is that pieces 2 and 3 need the *same* underlying oscillatory integral – both are built from the neutron-density residual, `profile_model.residual_integral_neutron(1a, 1b)` – and differ only in which potential-conversion function is applied to it afterwards (`matter_potential_nc` for piece 2, `matter_potential_cc` for piece 3, exactly matching Eq. (2.1)’s $V_{NC}(n_n)$ versus $V_{CC}(n_n)$) and which direction matrix they sandwich. `P_eps_n` therefore required *no new residual-integral machinery*: it reuses `residual_integral_neutron` exactly as `P_nc` already did.

Zeroth-order composition term and the neutron-coupling guard

`evolutor_perturbative_segment` computes `include_eps_n = oscillation.BSM_extension_NSI` and `oscillation.nsi.has_neutron_coupling` and, when true, builds the zeroth-order piece-3 contribution from `profile_model.average_n` (the segment-average neutron density) via `matter_potential_cc` – *not* from `profile_model.potential_n`, which is already converted with the *neutral-current* prefactor for piece 2’s own use and would apply the wrong prefactor if reused directly for piece 3. As in Section 2.3, a non-zero ϵ^n with a `profile_model` that was not built with neutron-density coefficients (`average_n/potential_n/residual_integral_neutron` unavailable) raises `ValueError` rather than silently ignoring the request – mirroring, and reusing the same underlying neutron-density data as, the pre-existing `include_matter_nc` guard for piece 2.

2.5 Pipeline wiring: fetching neutron density for the right reason

Approximations and Limitations

The subtlety this section documents. Piece 2 (sterile NC) and piece 3 (NSI composition) both need local neutron-density data, but they are triggered by *independent* conditions: piece 2 by the sterile-specific, tri-state `include_matter_nc` policy (`core.common.oscillation.resolve_include_matter_nc`, unchanged by this extension), piece 3 automatically by `oscillation.nsi.has_neutron_coupling`, with no separate flag – exactly like ϵ itself needs no “enable NSI” flag beyond `oscillation.nsi` being set. Every `medium.*` pipeline that decides *whether to sample/fetch* raw neutron-density data at all (as opposed to whether to *apply* the sterile NC term, which is already independently gated downstream) must therefore OR the two conditions together, via the shared helper

```
def oscillation_needs_neutron_composition(oscillation):
    return oscillation.BSM_extension_NSI and oscillation.nsi.
        has_neutron_coupling
```

in `core.common.oscillation`.

Implementation in tpeanuts

- **Earth.** `EarthProfile` bakes its neutron-density coefficients in at *construction* time (`profile_perturbative_kwargs='include_neutron': True`), independently of any per-call flag – so the exact (`evolutor_numerical`-based) and perturbative (`evolutor_perturbative_segment`-based) Earth paths both automatically get composition-term support for free once the profile was built with neutron data, requiring *no* pipeline-level change. The one exception is `earth_probability_state_numerical`, which (unlike the perturbative Earth path) samples `n_n` itself from the profile and previously did so only when `include_matter_nc` was true; it now also samples it when `oscillation_needs_neutron_composition` is true, raising a dedicated `ValueError` if the profile lacks neutron data.
- **Atmosphere.** Both `atmosphere_evolver_numerical` and `atmosphere_evolver_analytical` decide whether to sample/-fit neutron density *per call* (unlike Earth); both now OR in `oscillation_needs_neutron_composition` alongside the resolved `include_matter_nc`.
- **Sun.** `solar_evolver_numerical_history` (feeding `core.numerical.evolver_numerical`) and `solar_probability_mass`'s `method="adiabatic_exact"` branch (feeding `mass_weights_adiabatic_exact`, which itself calls `hamiltonian_flavour`) both apply the same OR condition before sampling `medium.density_n`. A related, more subtle fix was needed inside `mass_weights_adiabatic_exact` itself: its fixed-permutation bookkeeping (relabelling `torch.linalg.eigh`'s ascending-eigenvalue column order by physical vacuum mass index, needed because that order disagrees with the physical one under inverted ordering) diagonalises an auxiliary *vacuum-limit* Hamiltonian at $n_e = n_n = 0$; that internal call used to pass `n_n_mol_cm3=None` (physically inert either way, since $V_{CC}(0) = 0$), which now trips the §2.3 guard whenever $\epsilon^n \neq 0$. It now passes an explicit zero tensor instead of `None` – numerically identical, but compatible with the new guard.

Approximations and Limitations

`method="adiabatic_approximated"` remains **Standard-Model only**. The closed-form two-level matter-mixing-angle formulas underlying that method have no NSI generalisation at all (not even for $\epsilon^n = 0$); NSI of either kind is rejected there with `ValueError` regardless of composition, forcing `method="adiabatic_exact"` or `method="numerical"` for any NSI scenario in the Sun.

2.6 Not yet supported

Approximations and Limitations

The composition term (piece 3) has *no* generalisation implemented for `medium.solar.adiabatic.mass_weights_adiabatic_approximated` or its Landau-Zener correction, for the same structural reason ϵ itself has none there (§2.5). It is fully implemented for the exact evolver path (any medium) and for the perturbative

segment evoluter (Earth, atmosphere); the only propagation *method* left without composition-term support is the closed-form adiabatic approximation, which was already NSI-incompatible before this extension.

2.7 Autograd: fitting ϵ by gradient descent

`inference/solar_model.py`

SolarNSIOscillationModel – a differentiable solar $P_{ee}(E)$ model with a free diagonal NSI coupling.

Implementation in tpeanuts

`SolarNSIOscillationModel.oscillation(theta)` builds a fresh `NSIConfig` directly from the free-parameter tensor,

```
nsi = NSIConfig(
    eps_ee=values["eps_ee"],          # a torch.Tensor, requires_grad=True
    device=self.context.device,
    real_dtype=self.context.dtype,
)
```

relying on [Section 2.1](#)'s differentiable `_hermitian_3x3` to keep `eps_ee`'s gradient connected all the way through `epsilon`, `hamiltonian_flavour`, and `mass_weights_adiabatic_exact`'s `torch.linalg.eigh` diagonalisation, to the predicted P_{ee} . This constructor call is what `NSIConfig.from_raw_epsilon` (a one-hot $\times$ `eps_ee` matrix multiplication, in an earlier version of this model) existed to work around before [Section 2.1](#)'s rewrite – the differentiable scalar-field path removes the need for that workaround for any coupling that decomposes into the ordinary `eps_*/eps*_n` parametrization, which a single diagonal coupling like `eps_ee` always does. `inference.fit.fit_lbfgs` then differentiates through the whole chain with `torch.optim.LBFGS`, exactly as it already does for the purely-Standard-Model free parameters (`theta12`, `DeltamSq21`, ...); see [Section 4.4](#) for the resulting fit.

2.8 Summary: modules touched by this extension

Module	Role
core/BSM/bsm_nsi.py	NSIConfig : parameters, differentiable construction, embedding
core/common/hamiltonian.py	hamiltonian_matter_reduced : the three-piece decomposition
core/common/oscillation.py	oscillation_needs_neutron_composition
core/numerical/evolutor.py	Exact matrix-exponential evolution; consumes n_n_mol_cm3 generically
core/perturbative/evolutor.py	evolutor_perturbative_segment : P_eps_n sandwich term
medium/earth/probability.py	Neutron-density sampling for earth_probability_state_numerical
medium/atmosphere/evolutor.py	Neutron-density sampling for both atmosphere propagation methods
medium/solar/evolutor.py	Neutron-density sampling for the numerical solar evolutor
medium/solar/probability.py	Neutron-density sampling for method="adiabatic_exact"
medium/solar/adiabatic.py	Vacuum-limit permutation fix in mass_weights_adiabatic_exact
inference/solar_model.py	Differentiable eps_ee fit via SolarNSIOscillationModel

Presets, Observables, and Simulations

3.1 Named NSI presets

`config/presets.py`

NSI_PRESETS: named, literature-justified ϵ configurations, built via `NSIConfig.from_preset`.

Implementation in tpeanuts

Every preset below sets a handful of ϵ parameters to a representative, literature-motivated value (all other components default to zero) rather than to a unique global best fit: current data leaves large NSI degeneracies, so no single preferred point exists. Their purpose is threefold: (1) give the validation notebooks and tests physically motivated ϵ values instead of arbitrary numbers, covering diagonal-only, off-diagonal, single- and multi-parameter regimes; (2) reproduce, and stay traceable to, specific published sensitivity/bound studies; and (3) let users reach a standard benchmark scenario (e.g. the LMA-Dark degeneracy) by name instead of re-deriving its ϵ values. **None of the nine presets currently sets any ϵ^n (neutron-normalized) component:** every one is, in the language of [Section 1.3](#), a single-composition fit with an implicit $\epsilon^n = 0$ hypothesis. A user who wants to explore composition dependence explicitly builds an `NSIConfig` with both blocks set directly, e.g. `NSIConfig(eps_ee=0.30, eps_ee_n=0.10)`, or starts from a preset and overrides it with `dataclasses.replace` (safe for a preset-built config, since it was constructed from the scalar fields, not `from_raw_epsilon` – [Section 2.1](#)).

Preset	ϵ values and origin
sm_no_nsi	All $\epsilon = 0$. Standard Model limit; equivalent to passing <code>nsi=None</code> .
nsi_diagonal_biggio2009	$\epsilon_{ee} = 0.30$, $\epsilon_{\tau\tau} = 0.15$. Illustrative diagonal-only benchmark below the one-parameter limits collected by Biggio, Blennow, and Fernandez-Martinez [2]; those limits depend on the operator, fermion and normalization conventions.
nsi_lma_dark_esteban2018	$\epsilon_{ee} = -2.0$. The LMA-Dark degenerate solar solution of Esteban et al. [5] (§1.5): this large negative ϵ_{ee} flips the sign of the effective MSW potential, mimicking $\theta_{12} \rightarrow \pi/2 - \theta_{12} \approx 56.6^\circ$ (paired with the "_LMA_DARK_NUFIT52_NO" oscillation preset). Validated numerically in Section 4.2.
nsi_offdiag_icecube2022	$\epsilon_{\mu\tau} = 0.005$, $\epsilon_{\tau\tau} = 0.015$. Illustrates the sector and parameter scale probed by the IceCube DeepCore atmospheric analysis [10]; this exact pair is not a published best fit or confidence-limit point.
nsi_dune_etau	$\epsilon_{e\tau} = 0.15$ (real). Illustrative DUNE propagation-NSI point motivated by the octant-degeneracy study of Agarwalla, Chatterjee, and Palazzo [1]; it is not a DUNE best-fit point.
nsi_hyperk_etau	$\epsilon_{ee} = 0.20$, $\epsilon_{e\tau} = 0.10$ (real). Illustrative combination of a diagonal MSW correction and a flavour-changing coupling, motivated by Hyper-Kamiokande NSI-sensitivity studies [7, 11]; the exact pair is not a published best fit.
nsi_globalfit_esteban2018	$\epsilon_{ee} = 0.25$, $\epsilon_{e\tau} = 0.12$, $\epsilon_{\mu\tau} = 0.01$, $\epsilon_{\tau\tau} = 0.08$. Multi-parameter illustrative stress test inspired by the global oscillation analysis of Esteban et al. [5]; it is not a published fit point or an allowed-region claim, since correlations, degeneracies, and matter composition all matter (§1.3).
nsi_ee_only	$\epsilon_{ee} = 0.30$. Simplest possible NSI scenario: only the effective MSW potential is modified, with no flavour-changing coupling [2].
nsi_mutau_only	$\epsilon_{\mu\tau} = 0.01$ (real). Illustrative atmospheric stress-test just outside the IceCube DeepCore 90% CL interval $-0.0067 < \epsilon_{\mu\tau} < 0.0081$ [9]; it is not an allowed-point claim.

Key References

[1, 2, 5, 7, 9–11].

3.2 Building an oscillation scenario with NSI

Attaching an NSI preset to a propagation config

```
from tpeanuts.config.propagation import PropagationConfig
from tpeanuts.util.context import RuntimeContext

context = RuntimeContext.resolve("cpu", torch.float64)
oscillation = PropagationConfig.oscillation_parameters_from_preset(
    "_SM_NUFIT52_NO",
    NSI_extension="nsi_diagonal_biggio2009",  # any name in NSI_PRESETS
    context=context,
```



```
)
# oscillation.nsi is now a fully populated NSIConfig; every downstream
# builder (hamiltonian_reduced/flavour, evolver_numerical,
# evolver_perturbative_segment, medium.*) reads it automatically.
```

3.3 Observables and simulations that can be built with NSI

Implementation in tpeanuts

Every observable `tpeanuts` exposes for the Standard Model can, in principle, be re-evaluated with `oscillation.nsi` set, subject to the method-compatibility notes of §2.5:

- **Solar** $P_{ee}(E)$. `medium.solar.probability.solar_probability_state/solar_probability_mass` with `method="adiabatic_exact"` (pointwise Hamiltonian diagonalisation, any NSI ϵ , and, since this extension, ϵ^n whenever `medium.density_n` is available) or `method="numerical"` (full coherent matrix-exponential propagation through the tabulated solar density profile). Used for the LMA/LMA-Dark degeneracy study (Section 4.2) and the Borexino NSI inference fit (Section 4.4).
- **Earth regeneration and the day–night asymmetry.** `medium.earth.probability.earth_probability_state/_numerical/_analytical`, propagating a solar (or any other) incident state through the Earth’s PREM-based density profile. NSI in the Earth modifies the regeneration probability and hence the day–night asymmetry A_{DN} – the observable used to disfavour LMA-Dark independently of the Sun (Section 4.2).
- **Atmospheric oscillograms.** `medium.atmosphere.probability.atmosphere_probability_transition` and the combined `pipeline.atmosphere_earth` entry points, for a full atmosphere-production-to-underground-detector NSI oscillogram in $(E, \cos \theta_z)$.
- **Gradient-based inference.** `inference.solar_model.SolarNSIOscillationModel` (Section 2.7) fits a diagonal NSI coupling jointly with standard oscillation parameters by gradient descent against real solar $P_{ee}(E)$ data points, exactly as `SolarSMOscillationModel` does for the pure Standard Model.

Approximations and Limitations

As documented in Section 2.5, `method="adiabatic_approximated"` (the closed-form two-level solar approximation and its Landau-Zener correction) rejects any NSI configuration outright, and no propagation method currently generalises the composition term to `mass_weights_adiabatic_approximated`.

Notebook Results

This chapter reports the actual results produced by the notebooks that exercise NSI: two physics-validation notebooks (`notebooks/validation/physics/BSM/NSI1_test.ipynb`, `NSI2_lma_dark.ipynb`), one cross-method intrinsic-validation notebook (`notebooks/validation/intrinsic/intrinsic2_BSM_extensions.ipynb`) and one real-data inference notebook (`notebooks/inference/inference2_borexino_nsi.ipynb`).

Approximations and Limitations

On the completeness of the reported numbers. Several of these notebooks' saved `.ipynb` files have code cells with `execution_count: null` and an empty output list, even though later cells or their own markdown narrative reference a result – almost certainly because the file was saved from a session whose full re-run output was not re-serialized for every cell. This chapter states explicitly, for every quoted number, whether it is a value directly recorded in the notebook's saved output (**confirmed**) or a value only implied by later cells/markdown prose without its own recorded output (**unconfirmed, not independently verifiable from the file as saved**). No number is stated without this distinction. This is a bookkeeping limitation of the notebook files as currently saved, not a defect in the underlying `tpeanuts` computation itself, which [Chapter 2](#) documents and this project's automated test suite (`core/BSM/test/test2_bsm_nsi.py` and related) independently exercises on every change.

4.1 `NSI1_test.ipynb`: NSI deformation of the solar MSW resonance

Implementation in `tpeanuts`

This notebook builds the solar propagation Hamiltonian directly via `hamiltonian_flavour` for a battery of NSI presets, diagonalises it with `torch.linalg.eigh`, and integrates over the solar production profile – explicitly described in the notebook as "the same approach used internally by `solar_probability_state` when `epsilon` is provided." Presets exercised in its main results section: `sm_no_nsi`, `nsi_diagonal_biggio2009`, `nsi_offdiag_icecube2022`, `nsi_dune_etau`, `nsi_hyperk_etau`, and `nsi_globalfit_esteban2018` ([Section 3.1](#)).

Result

Confirmed, directly printed output:

SM closure error ($\epsilon = 0$ vs. analytic) = 3.879×10^{-6}

ΔP_{ee} at $E = 10$ MeV, relative to SM:

```
nsi_diagonal_biggio2009 : -0.0121
nsi_offdiag_icecube2022 : +0.0000
    nsi_dune_etau : -0.0201
    nsi_hyperk_etau : -0.0202
nsi_globalfit_esteban2018 : -0.0228
```

Diagonal ϵ_{ee} scan, 30 points in $[-0.10, 0.90]$:

```
max  $P_{ee}$  in scan = 0.5248
min  $P_{ee}$  in scan = 0.2910
```

The SM closure error of 3.9×10^{-6} compares the fully numerical $\epsilon = 0$ Hamiltonian-diagonalisation path against the closed-form adiabatic-approximated analytic path; the notebook's own interpretation is that this is "the intrinsic accuracy of the analytic path's mixing-angle factorisation... not floating-point-level" – i.e. the two are independent implementations of the same physics agreeing to five significant figures, not the same code path compared to itself.

Approximations and Limitations

The $\epsilon_{\mu\tau}$ -only preset (nsi_offdiag_icecube2022) produces $\Delta P_{ee} = 0.0000$ at 10 MeV to the precision printed: a purely $\mu\tau$ off-diagonal coupling has no first-order effect on ν_e survival at this energy/baseline within the printed precision, consistent with ν_e appearing only in the $ee/e\mu/e\tau$ entries of ϵ , none of which this preset touches.

The notebook produces four figure sets (sn5_fig3–sn5_fig6): an SM-validation panel ($P_{ee}(E)$ overlay plus the closure-error semilog plot above), a multi-preset $P_{ee}(E)$ comparison with a ΔP_{ee} panel, a 2-D heatmap of $P_{ee}(E, \epsilon_{ee})$ over the diagonal scan plus 1-D slices, and an off-diagonal $\epsilon_{e\mu}/\epsilon_{e\tau}$ scan. The notebook's stated physical reading of these figures: diagonal ϵ_{ee} "simply rescales the effective matter potential...shifting the resonance energy without creating new crossings," while off-diagonal entries "introduce new level crossings...that have no SM analogue," and "current constraints ($|\epsilon_{e\alpha}| \lesssim 0.3\text{--}0.5$) allow $\mathcal{O}(5\text{--}15\%)$ shifts in P_{ee} " – consistent with the confirmed ΔP_{ee} values above, all in the few-percent range for $|\epsilon| \sim 0.1\text{--}0.3$ couplings.

4.2 NSI2_lma_dark.ipynb: the LMA-Dark degeneracy**Implementation in tpeanuts**

This notebook validates the LMA-Dark degeneracy of §1.5 end to end: the analytic $\theta_{12} \leftrightarrow \epsilon_{ee}$ degeneracy identity, the daytime solar $P_{ee}(E)$ curves via `solar_probability_state`, and – the observable that actually breaks the degeneracy – the Earth day–night asymmetry A_{DN} via `earth_probability_state_numerical` (which the notebook notes supports NSI directly through its `epsilon=...` argument, unlike the daytime solar pipeline used

here, which does not yet include NSI inside the solar interior for this specific comparison).

Result

Confirmed, directly printed output (configuration/degeneracy check):

$$\begin{aligned}\theta_{12}^{\text{SM}} &= 33.4100^\circ, & \theta_{12}^{\text{Dark}} &= 56.5900^\circ \\ \sin^2 \theta_{12}^{\text{SM}} &= 0.3032, & \cos^2 \theta_{12}^{\text{Dark}} &= 0.3032 \quad (\text{must match – confirmed match}) \\ \epsilon_{ee}^{\text{SM}} &= 0.0, & \epsilon_{ee}^{\text{Dark}} &= -2.0\end{aligned}$$

Confirmed, directly printed output (day–night asymmetry at $E = 10$ MeV, linearly interpolated on the Earth-propagation energy grid):

Scenario	A_{DN}	Interpretation printed by the notebook
SM LMA	+4.84%	expected > 0
SM + NSI Earth	−1.59%	sign flipped
LMA-Dark approximation	−2.85%	LMA-Dark signature (< 0)
Super-Kamiokande (external)	$+3.3 \pm 1.0\%$	favours LMA, disfavours LMA-Dark at $\sim 3\sigma$

Approximations and Limitations

The confirmed `tpeanuts` SM-LMA value, $A_{DN} = +4.84\%$ at 10 MeV, is numerically above the quoted Super-Kamiokande measurement, $+3.3 \pm 1.0\%$ [5]: both share the same sign (qualitatively consistent, both > 0), but the notebook itself does not discuss the ~ 1.5 percentage-point quantitative gap. Super-K's A_{DN} is a flux-averaged, detector-specific measurement, whereas the $+4.84\%$ figure here is evaluated at a single energy and nadir angle (30° , 1 km detector depth) without folding a detector response or a flux/exposure average, so the two are not expected to agree in detail; this is a genuine open comparison for anyone extending this notebook, not a resolved validation.

The notebook produces three figures: MSW eigenvalue curves $\lambda_\pm(n_e)$ for SM-LMA versus LMA-Dark, showing (per its own caption) "an anti-crossing visible inside the solar-core band" for SM-LMA and "no anti-crossing" for LMA-Dark ("LMA-Dark avoids the solar resonance in normal matter"); a daytime $P_{ee}(E)$ comparison annotated with real Borexino (pp , ${}^7\text{Be}$, pep) and SNO (${}^8\text{B}$) data points (used only as plot annotations here, not fitted – the fit itself is [Section 4.4](#)); and the $A_{DN}(E)$ comparison plot with the Super-K band. The notebook's own stated conclusion: "Super-K's $A_{DN} = +3.3 \pm 1.0\% > 0$ disfavours LMA-Dark at $\sim 3\sigma$," while explicitly caveating that "NSI in the solar interior (daytime degeneracy) is not yet implemented in `solar_probability_state`. The dark-angular daytime curve is therefore not the true LMA-Dark prediction but its first-octant approximation."

4.3 intrinsic2_BSM_extension_NSI.ipynb: cross-method validation

Implementation in tpeanuts

This notebook cross-validates the numerical (matrix-exponential) and perturbative/analytical propagation paths against each other for an NSI-active scenario (oscillation pre-set "_SM_NUFIT52_N0" with `NSI_extension="nsi_globalfit_esteban2018"`), across Earth propagation and atmosphere-to-surface propagation, with an absolute-error acceptance threshold of 10^{-3} .

Result

Confirmed, directly printed output (only two of this notebook's comparison cells have recorded numeric output in the saved file; see the caveat box above):

Pure flavour state through the Earth ($E \in [100, 2 \times 10^4]$ MeV $\times$ 90 log-spaced points, $\eta \in [0.05, 1.50]$ rad $\times$ 90 points; `method="numerical"`, 400 midpoint steps, vs. `method="analytical"`):

Quantity	Value
max. absolute error	2.770×10^{-2}
mean absolute error	7.026×10^{-4}
max. relative error	5.648×10^{-1}
mean relative error	5.429×10^{-3}
comparison points	71 523
absolute threshold	1.000×10^{-3}

Atmosphere production to surface ($E \in [10^2, 10^5]$ MeV $\times$ 90, $\theta \in [0^\circ, 180^\circ] \times 90$; `method="numerical"`, 600 steps, vs. `method="analytical"`, 6 cubic segments):

Quantity	Value
max. absolute error	1.651×10^{-11}
mean absolute error	2.198×10^{-13}
max. relative error	1.551×10^{-9}
mean relative error	1.616×10^{-11}
comparison points	43 874
absolute threshold	1.000×10^{-3}

Approximations and Limitations

The Earth-crossing case's maximum *relative* error, 0.5648, looks large in isolation, but the notebook's own methodology (its Section 0.3) applies relative-error comparison only where the reference probability exceeds 10^{-5} , precisely to avoid exactly this kind of division-by-small-number inflation; the maximum *absolute* error, 2.77×10^{-2} , is the physically meaningful figure and is the one the acceptance criterion is actually built around. Both confirmed comparisons pass a reasonable reading of the 10^{-3} absolute-error

criterion (the atmosphere case by many orders of magnitude, reflecting the atmosphere's much lower electron density; the Earth case's mean absolute error, 7.0×10^{-4} , is below threshold, while its *maximum* absolute error, 2.77×10^{-2} , exceeds it at the grid's most extreme, longest-baseline points – consistent with the first-order perturbative expansion's known degradation for long, high-density segments rather than an NSI-specific defect).

The notebook's design also includes five further comparisons (SM-limit closure, solar-surface `adiabatic_exact` vs. numerical, solar-to-detector, atmosphere-to-detector, and a final `assert np.isfinite(...)` summary printing "All four physical comparisons completed successfully.") whose cells have no recorded output in the saved file, so their outcome cannot be confirmed from the notebook artifact as saved; re-running the notebook would be needed to obtain them.

4.4 inference2_borexino_nsi.ipynb: fitting the degeneracy to real data

Implementation in tpeanuts

This is the one NSI notebook that fits real data: it performs a gradient-based, joint $(\theta_{12}, \epsilon_{ee})$ fit (`SolarNSIOscillationModel`, Section 2.7, via `inference.fit.fit_lbfgs`) to the four real Borexino $P_{ee}(E)$ points published in [5]-style analyses (loaded from `data/solar/borexino/probability/nature2018_pee_points.csv`), starting from two deliberately different initial guesses – one on the standard-LMA side ($\theta_{12} = 25^\circ$, $\epsilon_{ee} = +0.5$) and one on the LMA-Dark side ($\theta_{12} = 65^\circ$, $\epsilon_{ee} = -1.2$) – to test whether both converge to the degenerate solutions predicted by §1.5's analytic transformation. The base oscillation preset used is `"_SM_NUFIT61_NO"`.

Result

Confirmed, directly printed output – the degeneracy cross-check comparing the dark-side LBFGS fit against the value predicted by applying $\theta_{12} \rightarrow 90^\circ - \theta_{12}$, $\epsilon_{ee} \rightarrow -(2 + \epsilon_{ee})$ to the independently-converged LMA-side fit:

Quantity	Predicted from LMA fit	Actual dark-side fit	Difference
$\theta_{12}^{\text{dark}}$ [deg]	55.50	55.50	+0.00
$\epsilon_{ee}^{\text{dark}}$	-1.775	-1.782	+0.007

Both printed as within 1σ of the predicted value (`theta12_dark` within 1 sigma of prediction: `True`, `eps_ee_dark` within 1 sigma of prediction: `True`).

Confirmed, directly printed output – the independent 2-D $\Delta(-2 \ln \mathcal{L})$ grid scan (LMA-side branch, $30 \times 30 = 900$ points, $\theta_{12} \in [5^\circ, 85^\circ]$, $\epsilon_{ee} \in [-3.0, 1.0]$):

$$\chi_{\min}^2 = 0.777 \quad \text{at} \quad \theta_{12} = 35.34^\circ, \quad \epsilon_{ee} = -0.24 \quad (4 \text{ data points}, 2 \text{ free parameters}).$$

Approximations and Limitations

The exact numerical values of the LBFGS fits themselves (Section 4 of the notebook, including its own `assert delta_chi2 < 1.0` and "PASSED" message) have no recorded cell output in the saved file, so they are **not independently confirmable** from the artifact as saved. They can, however, be partially back-derived from the confirmed Section 5 table above: since $\theta_{12}^{\text{dark, predicted}} = 90^\circ - \theta_{12}^{\text{LMA}} = 55.50^\circ$, the LMA-side fit must have converged to $\theta_{12}^{\text{LMA}} \approx 34.50^\circ$; since $\epsilon_{ee}^{\text{dark, predicted}} = -(2 + \epsilon_{ee}^{\text{LMA}}) = -1.775$, $\epsilon_{ee}^{\text{LMA}} \approx -0.225$. These derived values are reasonably consistent with the independently confirmed grid-scan minimum, $\theta_{12} = 35.34^\circ$, $\epsilon_{ee} = -0.24$ (a coarse 30×30 grid evaluated independently of the gradient fit), which is expected agreement between two different optimisation methods rather than a coincidence, but this reasoning is presented here as *derived*, not as a number the notebook itself printed.

The notebook produces three figures: a filled $\chi^2(\theta_{12}, \epsilon_{ee})$ contour map over the full grid with the standard-LMA and LMA-Dark preset points marked as reference stars; an overlay of the two best-fit $P_{ee}(E)$ curves against the real Borexino data points with a residual-difference panel; and the $\Delta(-2 \ln \mathcal{L})$ uncertainty map with Wilks' 68%/90%/95% contours (at $\Delta\chi^2 = 2.30/4.61/6.18$ for 2 degrees of freedom) and the grid minimum marked. The notebook's own stated conclusion: "two deliberately different, uninformative starting guesses converge to two well-separated points...with statistically indistinguishable χ^2 ...quantitatively consistent with the $\theta_{12} \rightarrow 90^\circ - \theta_{12}$, $\epsilon_{ee} \rightarrow -(2 + \epsilon_{ee})$ relation," and "the two fitted $P_{ee}(E)$ curves agree to a fraction of a percent everywhere from 0.15 to 16 MeV" – but also, echoing Section 4.2, that "a real resolution of LMA vs. LMA-Dark needs independent information...not more solar $P_{ee}(E)$ points alone" (i.e. the Earth day–night asymmetry of Section 4.2, or another observable outside the solar $P_{ee}(E)$ degeneracy itself).

4.5 The composition-dependence extension: no notebook coverage yet

Approximations and Limitations

This is the most important caveat in this chapter. An exhaustive search of every notebook in the repository (grep for `eps_ee_n`, `eps_.*_n`, `epsilon_n`, `has_neutron_coupling`, and `composition`) confirms that **no notebook currently demonstrates the neutron-density composition-dependence extension** (ϵ^n , Sections 1.3 and 2.3). The handful of unrelated matches (an Earth/solar/AK135 raw-data column literally named `neutron_density_mol_cm3`, evolver-composition consistency checks unrelated to NSI, PMNS flavour-composition language, etc.) are all false positives on the search terms, not partial coverage. This is consistent with the extension's implementation history: it was added after every existing notebook was last executed, so no notebook artifact can yet be cited as numerical evidence for it. The automated test suite (`core/BSM/test/test2_bsm_nsi.py`, `core/common/test/test3_hamiltonian.py`, `core/perturbative/test/test2_perturbative_evolver.py`, and the `medium/earth`, `medium/atmosphere`, `medium/solar` test suites) is, at the time of writing, the only executed and verified evidence for the composition-dependence extension's correctness; a natural follow-up to this document would be a fifth notebook demonstrating it end to end – e.g. propagating a solar neutrino from the hydrogen-rich envelope ($n_n/n_e \rightarrow 0$) through the helium/metal core ($n_n/n_e \rightarrow 1$) with a non-zero ϵ^n , and comparing against the $\epsilon^n = 0$

approximation used throughout this chapter's existing notebooks.

References

- [1] S. K. Agarwalla, S. S. Chatterjee, and A. Palazzo. “Degeneracy between theta23 octant and neutrino non-standard interactions at DUNE”. In: *Phys. Lett. B* 762 (2016), pp. 64–71. eprint: [1607.01745](#).
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